E-COMMERCE FRAUD DETECTION USING MACHINE LEARNING

Project Documentation

1. Abstract

This project detects fraudulent e-commerce transactions using machine learning classification techniques. It analyzes transaction-related features to identify suspicious activities and assist organizations in reducing financial losses caused by fraud.

2. Problem Statement

Online transactions are increasing rapidly, making fraud detection difficult through manual inspection. An automated machine learning system is required to accurately classify fraudulent transactions.

3. Objectives

• Detect fraudulent transactions.
• Preprocess transaction data.
• Compare multiple machine learning models.
• Improve fraud detection accuracy.
• Support secure online payments.

4. Dataset Description

The project uses a transaction dataset (transactions.csv) containing transaction attributes and a target label indicating whether a transaction is fraudulent or legitimate.

5. Software Requirements

- **Programming Language:** Python
- **Development Environment:** Google Colab
- **Libraries Used:**
  - Pandas
  - NumPy

- Scikit-learn
- XGBoost
- LightGBM
- CatBoost
- Matplotlib
- Seaborn

6. Methodology

Step 1: Import libraries.
Step 2: Load the transaction dataset.
Step 3: Data preprocessing and feature preparation.
Step 4: Split data into training and testing sets.
Step 5: Train multiple classification models.
Step 6: Evaluate model performance.
Step 7: Select the best model for fraud prediction.

7. Exploratory Data Analysis (EDA)

Graph 1: Fraud Distribution – Displays the number of fraudulent and legitimate transactions.

Graph 2: Transaction Feature Distribution – Shows the spread of important transaction features.

Graph 3: Correlation Heatmap – Identifies relationships among transaction variables.

Graph 4: Model Accuracy Comparison – Compares the performance of all machine learning models.

8. Machine Learning Models Used

1. Logistic Regression

– Predicts fraudulent transactions using probability.

2. Decision Tree

– Creates decision rules for fraud classification.

3. XGBoost

– Improves predictions using boosted decision trees.

4. LightGBM

– Efficient gradient boosting model for large datasets.

5. CatBoost

– Handles categorical data effectively and improves prediction accuracy.

9. Model Evaluation

Models are evaluated using Accuracy Score, Classification Report and Confusion Matrix. The model with the highest performance is selected for final prediction.

10. Prediction Process

The trained model receives transaction details as input and predicts whether the transaction is fraudulent or legitimate.

11. Future Scope

- Real-time fraud monitoring.
- Cloud deployment.
- Explainable AI integration.
- Deep learning-based fraud detection.
- Banking and payment gateway integration.

12. Conclusion

The project demonstrates that machine learning can effectively identify fraudulent transactions, improving security and minimizing financial risk in e-commerce platforms.